Scratch Game Blueprint: "Catch the Falling Objects"

Build guide with sprite list, variables, and block logic for scratch.mit.edu

Overview

The player moves a basket left/right to catch falling collectibles (+10 points) while avoiding falling obstacles (-5 points, lose 1 life). The game has a start screen, a score, a minimum of 3 lives, a game over screen when lives reach 0, and a win message when score reaches 100.

1. Sprites and Backdrops

Create these in the Scratch editor:

- **Basket** — the player-controlled catcher (any basket/bowl sprite from the library)
- **Collectible** — the +10 point object (e.g. Apple, Star)
- **Obstacle** — the -5 point / lose-a-life object (e.g. Rock, Bomb)
- **Backdrop: Start Screen** — plain backdrop with game title text drawn on it, or use a Stage variable to show instructions
- **Backdrop: Game Backdrop** — the play area background

2. Variables (Data category → Make a Variable)

- **score** — for all sprites
- **lives** — for all sprites
- **game state** — for all sprites (values: "start", "playing", "gameover", "win")

3. Start Screen Logic (Stage)

```
when green flag clicked
switch backdrop to [Start Screen]
set [game state] to [start]
set [score] to 0
set [lives] to 3
broadcast [reset game]
wait until <mouse down?> or <key [space] pressed?>
switch backdrop to [Game Backdrop]
set [game state] to [playing]
broadcast [start game]
```

Add a text costume/stamp on the Start Screen backdrop that says the game title (e.g. "Catch the Falling Objects") and instructions ("Use arrow keys to move the basket. Press SPACE or click to start.").

4. Basket Sprite (player control)

```
when green flag clicked
go to x: 0 y: -140
show

when I receive [start game]
forever
```

```
    if <key [left arrow] pressed?> then
        change x by (-8)
    end
    if <key [right arrow] pressed?> then
        change x by (8)
    end
    if <(x position) < (-220)> then
        set x to (-220)
    end
    if <(x position) > (220)> then
        set x to (220)
    end
end
```

5. Collectible Sprite (+10 points)

```
when I receive [start game]
forever
    go to x: (pick random -220 to 220) y: (180)
    show
    repeat until <touching [Basket]?> or <(y position) < (-180)>
        change y by (-4)
    end
    if <touching [Basket]?> then
        change [score] by (10)
        play sound [collect]
        hide
        if <(score) >= (100)> then
            broadcast [win game]
        end
    end
end
```

6. Obstacle Sprite (-5 points, lose a life)

```
when I receive [start game]
forever
    go to x: (pick random -220 to 220) y: (180)
    show
    repeat until <touching [Basket]?> or <(y position) < (-180)>
        change y by (-5)
    end
    if <touching [Basket]?> then
        change [score] by (-5)
        change [lives] by (-1)
        play sound [hit]
        hide
        if <(lives) <= (0)> then
            broadcast [game over]
        end
    end
end
```

7. Game Over Screen (Stage)

```
when I receive [game over]
set [game state] to [gameover]
switch backdrop to [Game Over Screen]
stop [other scripts in sprite]
```

Add a "Game Over Screen" backdrop with a text costume showing "Game Over — Final Score: " plus a score variable display (checkbox the score variable to show it on stage).

8. Win Screen (Stage)

```
when I receive [win game]
set [game state] to [win]
switch backdrop to [Win Screen]
stop [other scripts in sprite]
```

Add a "Win Screen" backdrop with text like "You Win! Score reached 100!"

9. Stopping falling sprites on game end

Add this script to BOTH the Collectible and Obstacle sprites so they stop moving once the game ends:

```
when I receive [game over]
hide
stop [this script]

when I receive [win game]
hide
stop [this script]
```

10. Checklist Before Submitting

- Start screen shows the game title and instructions
- Basket moves left/right with arrow keys (or mouse) and stays on screen
- Collectible falls, gives +10 score on catch
- Obstacle falls, gives -5 score and -1 life on catch
- Score variable is visible and updates correctly
- Lives variable starts at 3 (or more) and updates correctly
- Game Over screen appears when lives reach 0
- Win message appears when score reaches 100
- Project is shared on scratch.mit.edu (click Share) and the link is submitted, or screenshots of the stage + block scripts are exported as a PDF